Joshua Ebinisi

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EDUCATION

University of Houston Houston, Texas

Bachelor of Science in **Computer Engineering,** Honors Program – GPA: **3.73** May 2028

Relevant coursework: Programming Fundamentals II, Physics, Computing for Engineers, Calculus I & II, Macroeconomics, AI/ML Concepts

Affiliations: NSBE, SEO Career, Codepath, IEEE

Honors: Dean's List, Academic Excellence Scholarship, Georgia CCE Honor Society, GGC President's List, DoE Intl. Award, Harvard YL Program

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, Bash, PowerShell, HTML/CSS

Frameworks/Libraries: PySpark, PyTorch, Pandas, NumPy, Seaborn, Matplotlib, Streamlit, React, Node.js, Pickle, Scikit-learn, yFinance, MySQL

Data Platforms: Hadoop, Hive, Impala, Spark

Tools/Platforms: Jupyter Notebook, VS Code, Obsidian, Git, TradingView, Slurm, Bitbucket, Azure DevOps, Microsoft Office Suites, MSAL, Alation, MATLAB, Arduino, Tinker CAD, Unity

EXPERIENCE

JPMorgan Chase New York City, NY

Data for Good Hackathon Participant October 2026

- Engineered a statistical model of how NBA travel and scheduling affect team performance across **6,150** games, finding back-to-backs had the largest effect, dropping teams to a **44%** win rate; collaborated in a diverse **4**-person team using Python, Jupyter Notebooks, Pandas, and statsmodels.

PNC Washington D.C.

Quantitative Analytics Intern June 2026 – August 2026

- Applied quantitative and data modeling techniques while collaborating on over **400** models across PNC's inventory, including a CCAR/DFAST loss-forecasting model developed through **1,000** Monte Carlo simulations, validating a **90**% prediction interval against actual quarterly results, deployed using Python's pickle library.
- Built and productionized an automated Python data pipeline for regulatory and risk analytics, monitoring internal and external sources to deliver daily reporting for Model Risk, MRMG, Audit, and OP Risk stakeholders, boosting model development output by **10**%.
- Engineered an end-to-end big data pipeline (Python, PySpark, Hadoop, Hive, Impala) processing **2M+** customer transactions to detect spending spikes and portfolio shifts to trigger targeted banker outreach, scheduled through Slurm/PowerShell automation, driving a **5**% increase in client retention.

Code Ninjas Snellville, Georgia

Coding Instructor January 2025 – January 2026

- Educated 30+ children (ages 5-15 years) in developing their skills in MS Office, JavaScript, and Unity, leading to increased interest in content creation, game design, and programming careers that led to a **10**% increase in summer enrollment during summer sessions.
- Provided individualized support on tasks and projects, driving a **25**% increase in Black Belt students and earning the branch the Black Belt Center Award for most skilled students across Georgia.

PROJECTS

AI-Powered Governance & Model Development News Bot | Python, JavaScript, HTML, CSS, Power Automate June 2026

- Developed an AI-powered news bot for PNC's Quant team that uses keyword-based filtering to surface the latest governance and model development updates in real time, boosting model development output by **10**%.
- Designed a chatbot interface allowing team members to query updates by specific governance or regulatory events on-demand, reducing manual research time and improving turnaround on time-sensitive changes.

Signal-Modulated DCA Backtester | Python, Pandas, NumPy, yFinance January 2026

- Built an automated backtester on the Dollar Cost Average strategy for Nasdaq 100 futures, layering RSI, volatility, and MA200 filters to scale monthly demo investing, achieving an 832% return using **75**% less capital than standard DCA.
- Verified the strategy's reliability by holding all rules fixed ($100 invested every 30 days) and testing it across **5** different starting years (2010–2018), where it outperformed a simple buy-and-hold approach by **12.4**% over a 15-year period.

Donchian Breakout Strategy: BTC/USD | Python, Pandas, NumPy, yFinance January 2026

- Engineered a Donchian Breakout trading algorithm in Python to demo trade BTC/USD, optimizing lookback windows across **6** years (2020-2026) of price data to generate systematic long or short signals.
- Backtested the strategy on $10,000 of starting capital, generating a 742% total return while cutting max drawdown by 40% compared to buy and hold through the 2022 crypto market downturn.

SortSense: The Interactive Recycling Challenge | HTML, CSS, JavaScript October 2025

- Developed an educational web-based game prototype in a **90**-minute hackathon to address local recycling contamination by using a tool to focus on improving user source separation across 10 common household items (Plastics, Papers, Cans).
- Integrated dynamic game logic with instant feedback and live leaderboard to encourage a friendly high-scoring competition in environment literacy, successfully demonstrating a solution that targets plastic pollution and is designed to achieve a **2**% growth in environmental sustainability on campus.

LEADERSHIP INVOLVEMENT

National Society of Black Engineers (NSBE) Houston, Texas

Co-Finance Chair August 2025 – Present

- Executed new fundraising events and managed chapter financial planning, resulting in a **5**% organizational growth for the academic year.

Streetwise Georgia Lawrenceville, Georgia

Student Volunteer July 2024 – June 2025

- Contributed over **60** hours to supporting **100**+ families by coordinating food and clothing distributions, warehouse operations, and seasonal events, effectively reducing their monthly living expenses by over $**200.**

Earth Savers Community Service Group Lagos, Nigeria

President March 2022 – September 2024

- Dedicated over **200** hours to community fundraising and the **₦1.5 million** renovation of Agbara Community High School in collaboration with Society Developers, successfully allowing us to achieve UNICEF Sustainable Development Goals **3** and **4**.